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CHARGE PUMP WITH PASS TRANSISTOR CONTROLLED BY A SUPPLEMENTARY CHARGE PUMP STAGE

Abstract

Disclosed is a charge pump circuit including: a single primary charge pump stage or multiple primary charge pump stages connected in series; a pass transistor connected between an output node of the primary charge pump stage (or of the last of the multiple primary charge pump stages, if applicable) and an output terminal; and a supplementary charge pump stage. The supplementary charge pump stage receives the same input voltage as the primary charge pump stage (or the same input voltage as the last of the multiple primary charge pump stages, if applicable) and controls the gate of the pass transistor to reduce ripple at the output terminal. A capacitive load can also be connected to the output terminal to reduce ripple and the size of the capacitive load can be selected to achieve a desired balance between the amount of ripple and circuit size.

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Background/Summary

BACKGROUND

[0001] The present disclosure relates to charge pumps and, more particularly, to embodiments of a charge pump configured for reduced output voltage ripple and/or area.

[0002] A charge pump circuit is a circuit configured to convert a direct current (DC) power source (i.e., an input voltage (Vin)) to a larger DC power source (e.g., to an output voltage (Vout) that is greater than Vin). The circuit can be a single-stage charge pump circuit or a multi-stage charge pump circuit. A single-stage charge pump circuit can, for example, convert a Vin that is equal to a positive supply voltage (VDD) to a Vout that is approximately 2*VDD or somewhat less when an electrical load is connected to the output to drive current. For example, if Vin is 1.8V, then Vout could be approximately 3.6V or reduced to, for example, 3.0V due to a resistive load connected to the output. A multi-stage charge pump can include n-stages and can, for example, convert a Vin that is equal to VDD to a Vout that is approximately (n+1)*VDD or somewhat less when an electrical load is connected to the output to driver current. Regardless of whether the charge pump is a single-stage or a multi-stage charge pump circuit, Vout may exhibit significant ripple. Ripple refers to periodic variation exhibited by a DC voltage and this variation can negatively impact performance of devices connected to receive Vout. Typically, Vout ripple is reduced by increasing the capacitance load (CL) at the output of the charge pump. However, adding capacitance can significantly increase circuit area.

SUMMARY

[0003] Disclosed herein are embodiments of a structure and, particularly, of a charge pump circuit.

[0004] In some embodiments, the structure can include a primary charge pump stage with an input node and an output node. The structure can further include a supplementary charge pump stage with an additional input node and an additional output node. The input node of the primary charge pump stage and the additional input node of the supplementary charge pump stage can be connected to receive an input voltage. The structure can further include an output terminal and a pass transistor, which is connected between the output terminal and the output node of the primary charge pump and which has a gate connected to the additional output node of the supplementary charge pump stage.

[0005] In other embodiments, the structure can include a primary charge pump stage with an input node and an output node. The structure can further include a supplementary charge pump stage with an additional input node and an additional output node. The input node of the primary charge pump and the additional input node of the supplementary charge pump can be connected to receive an input voltage. The structure can further include an output terminal, which is connected to a capacitive load. The structure can further include a pass transistor, which is connected between the output terminal and the output node of the primary charge pump stage and which has a gate connected to the additional output node of the supplementary charge pump stage.

[0006] In still other embodiments, the structure can include multiple primary charge pump stages connected in series. The primary charge pump stages can include at least a first primary charge pump stage and a last primary charge pump stage. Furthermore, the last primary charge pump stage can include a last input node and a last output node. The structure can further include a supplementary charge pump stage with an additional input node and an additional output node. The last input node of the last charge pump stage and the additional input node of the supplementary charge pump stage can be connected to receive a last input voltage from a preceding primary charge pump stage. The structure can further include an output terminal and a pass transistor, which is connected between the output terminal and the last output node of the last primary charge pump

stage and which has a gate connected to the additional output node of the supplementary charge pump stage.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The present disclosure will be better understood from the following detailed description with reference to the drawings, which are not necessarily drawn to scale and in which:

[0008] FIG. **1**A is a schematic diagram illustrating disclosed embodiments of a charge pump circuit;

[0009] FIG. **1**B is a schematic diagram illustrating an example of a clock driver that can be incorporated into the charge pump circuit of FIG. **1**A;

[0010] FIGS. **2**A and **2**B are graphs illustrating and comparing example values for PCP-Vout and for SCP-Vout in the charge pump circuit of FIG. **1**A;

[0011] FIG. **2**C is a graph illustrating a Vout in one example where reducing ripple is prioritized over reducing area;

[0012] FIG. **2**D is a graph illustrating a Vout in another example where reducing area is prioritized over reducing ripple; and

[0013] FIG. **3** is a schematic diagram illustrating additional disclosed embodiments of a charge pump circuit.

DETAILED DESCRIPTION

[0014] As mentioned above, a charge pump circuit is a circuit configured to convert a direct current (DC) power source (i.e., an input voltage (Vin)) to a larger DC power source (e.g., to an output voltage (Vout) that is greater than Vin). The circuit can be a single-stage charge pump circuit or a multi-stage charge pump circuit. A single-stage charge pump circuit can, for example, convert a Vin that is equal to a positive supply voltage (VDD) to a Vout that is approximately 2*VDD or somewhat less when an electrical load is connected to the output to drive current. For example, if Vin is 1.8V, then Vout could be approximately 3.6V or reduced to, for example, 3.0V due to a resistive load connected to the output. A multi-stage charge pump can include n-stages and can, for example, convert a Vin that is equal to VDD to a Vout that is approximately (n+1)*VDD or somewhat less when an electrical load is connected to the output to driver current. Regardless of whether the charge pump is a single-stage or a multi-stage charge pump circuit, Vout may exhibit significant ripple. Ripple refers to periodic variation exhibited by a DC voltage and this variation can impair performance of devices connected to receive Vout. Typically, Vout ripple is reduced by increasing the capacitance load (CL) at the output of the charge pump. However, adding capacitance can significantly increase circuit area.

[0015] In view of the foregoing, disclosed herein are embodiments of a structure and, particularly, a charge pump circuit configured for reduced output voltage ripple at the output terminal of the circuit and/or for reduced circuit area. In each of the embodiments, the circuit can include a single primary charge pump stage or multiple primary charge pump stages connected in series. A pass transistor can be connected between an output node of the primary charge pump stage (or an output node of the last of the multiple primary charge pump stages, if applicable) and the output terminal of the circuit. A supplementary charge pump stage can be connected to receive the same input voltage as the primary charge pump stage (or the same input voltage as the last of the multiple primary charge pump stages, if applicable) and can further control the gate of the pass transistor in order to reduce the output voltage ripple at the output terminal. The circuit can further include a capacitive load connected to the output terminal, also to reduce output voltage ripple. Thus, during design, the size of the capacitive load can be selected to achieve a desired balance between the amount of output voltage ripple and the overall size of the circuit (which increases or decreases as a function of the load capacitance).

[0016] More particularly, referring to FIG. **1**A, disclosed herein are embodiments of a structure and, particularly, of a charge pump circuit **100** (hereinafter referred to as circuit **100**) including a single primary charge pump (PCP) stage **110**.

[0017] PCP stage **110** can include an input node **115**, an output node **116**, and a pair of cross-coupled inverters **111** and **112** connected between input node **115** and output node **116**. One inverter **111** can include an N-channel field effect transistor (NFET) **111***a* and a P-channel field effect transistor (PFET) **111***b* connected in series between input node **115** and output node **116**. Another inverter **112** can similarly include an NFET **112***a* and a PFET **112***b* connected in series between input node **115** and output node **116**. An intermediate node **117** at the junction between NFET **111***a* and PFET **111***b* (e.g., on an interconnect electrically connecting NFET **111***a* to PFET **111***b*) can be electrically connected to the gates of NFET **112***a* and PFET **112***b*. Similarly, an intermediate node **118** at the junction between NFET **112***a* and PFET **112***b* (e.g., on an interconnect electrically connecting NFET **112***a* and PFET **112***b*) can be electrically connected to the gates of NFET **111***a* and PFET **111***b*.

[0018] PCP stage **110** can further include a pair of clock signal nodes **121** and **122**. Clock signal nodes **121** and **122** can be electrically connected to receive a clock signal (CLK**1**) and an inverted clock signal (CLK**1**B), respectively, from a clock driver **105**, as discussed in greater detail below.

[0019] PCP stage **110** can further include a pair of capacitors **113** and **114** connected between the clock signal nodes **121** and **122** and the intermediate nodes **117** and **118**, respectively. That is, capacitor **113** can include conductive plates, which are connected to clock signal node **121** and intermediate node **117**, respectively, and which are separated from each other by a capacitor dielectric. Capacitor **114** can similarly include conductive plates, which are connected to clock signal node **122** and intermediate node **118**, respectively, and which are also separated from each other by a capacitor dielectric.

[0020] In this PCP stage **110**, input node **115** can be connected to receive a direct current (DC) power source (i.e., an input voltage (Vin)). Vin can, for example, be at the level of a relatively low positive supply voltage (VDDA). This relatively low VDDA can, for example, correspond to the maximum voltage ratings of the devices used (e.g., of NFET **111***a*, PFET **111***b*, NFET **112***a*, and PFET **112***b*). For example, if NFET **111***a*, PFET **112***b*, NFET **112***a*, and PFET **112***b* are all rated as 1.8 volt (V)-FETs (e.g., with a maximum gate-source voltage (VGS) rating of 1.8V, a maximum gate-drain voltage (VDS) of 1.8V, and a maximum source-drain voltage (VSD) rating of 1.8V), then VDDA could be 1.8 volts (V). If NFET **111***a*, PFET **111***b*, NFET **112***a*, and PFET **112***b* are all 1.5V-FETs, then VDDA could be 1.5V, and so on.

[0021] Additionally, in this PCP stage **110**, CLK**1** can swing periodically between 0.0V and VDDA and CLK**1**B can swing periodically between VDDA and 0.0V. That is, as CLK**1** changes from VDDA to 0.0V, CLK**1**B switches from 0.0V to VDDA and vice versa. As mentioned above, CLK**1** and CLK**1**B can be received by PCP stage **110** from a clock driver **105**.

[0022] FIG. **1**B is a schematic diagram illustrating one example of a clock driver **105** that can be employed to output CLK**1** and CLK**1**B. As illustrated, clock driver **105** can include, for example, a NAND gate **101**, which is electrically connected to a positive power supply at VDDA and further electrically connected to GND (e.g., at 0.0V). NAND gate **101** can receive, as inputs, an initial clock signal (CLK), which swings between 0.0V and VDDA, and an enable signal (EN). NAND gate **101** can output a given logic value according to a conventional NAND gate truth table. That is, the logic value output by NAND gate **101** will be high (at VDDA) when the following conditions are met: CLK and EN are both low, CLK is low and EN is high, and CLK is high and EN is low. The logic value output by NAND gate **101** will be low (at 0.0V) when CLK and EN are both high. In operation, the logic value output by NAND gate **101** can oscillate between GND and VDDA (as a function of CLK and EN) effectively creating a voltage pulse. For purposes of this disclosure, a “voltage pulse” refers to an essentially rectangular direct current voltage signal that transitions between a low voltage level and a high voltage level at regular and repeated intervals such that the pulse is at the low voltage level for a first time period (t1), switches to the high voltage level and remains at the high voltage level for a second time period (t2) (which is the same as or different from t1), switches back to the low voltage level and again remains at the low voltage level for t1, and so on.

[0023] Clock driver **105** can further include a single inverter **102** (also referred to herein as an inverting buffer) electrically connected to receive the voltage pulse from NAND gate **101**. Clock driver **105** can further include a pair of series-connected inverters **103-104** (also referred to herein as series-connected inverting buffers) also electrically connected to receive the voltage pulse from NAND gate **101**. These inverters **102-104** can each be connected to the positive power supply at VDDA and to GND. In response to the voltage pulse from the NAND gate **101**, inverter **102** can output CLK**1** (which transitions between GND and VDDA) and the series-connected inverters **103-104** can output CLK**1**B (which is inverted with respect to CLK**1** so that when CLK**1** transitions from GND to VDDA, CLK**1**B transitions from VDDA to GND and vice versa). It should be noted that the example of the clock driver **105** described above is provided for illustration purposes and not intended to be limiting. Alternatively, any other suitable clock driver configured to generate and output CLK**1** and CLK**1**B, as described could be employed.

[0024] Referring again to FIG. **1**A, in operation, input node **115** of circuit **100** receives Vin, which is steady at VDDA. Clock signal node **121** and thereby capacitor **113** receives CLK**1**. Clock signal node **122** (and thereby capacitor **114**) receives CLK**1**B. As a result, the voltage signal at intermediate node **117** between NFET **111***a* and PFET **111***b* of inverter **111** transitions between VDDA and a higher voltage that is approximately double the magnitude of VDDA (e.g., +/−10%). Furthermore, the voltage signal at intermediate node **118** between NFET **112***a* and PFET **112***b* of inverter **112** will be inverted with respect to the voltage signal at intermediate node **117**. Thus, when CLK**1** switches to GND and CLK**1**B switches to VDDA, the voltage signal at intermediate node **117** drops to VDDA turning off NFET **112***a* and turning on PFET **112***b* and the voltage signal at intermediate node **118** is raised to 2*VDDA, turning on NFET **111***a* and turning off PFET **111***b*. Contrarily, when CLK**1** switches to VDDA and CLK**1**B switches to GND, the voltage signal at intermediate node **117** rises to 2*VDDA, turning on NFET **112***a* and turning off PFET **112***b* and the voltage signal at intermediate node **118** drops to VDDA turning off NFET **111***a* and turning on PFET **111***b*. As a result, the output voltage of PCP (PCP-Vout **119**) at output node **116** is continuously pulled toward 2*VDDA, regardless of the states of CLK**1** and CLK**1**B. Thus, for example, if VDDA is 1.8V, then PCP-Vout **119** would be continuously pulled toward 3.6V. Output node **116** of PCP stage **110** is further electrically connectable to an output terminal **185** of circuit **100**.

[0025] Optionally, circuit **100** can also include a resistive load **182**, which is electrically connected to output terminal **185** for driving current. This resistive load **182** may reduce the voltage level at output terminal **185** somewhat. For example, given a resistive load **182**, an adjusted output voltage (aVout) **190** at output terminal **185** may be pulled down somewhat. For example, if PCP-Vout **119** on output node **116** is on average at a voltage level of 2*VDDA (e.g., at 3.6V), then a Vout **190** at output terminal **185** may be average 600 mV less (e.g., 3.0V).

[0026] Furthermore, ripple of PCP-Vout **119** may be significant. As mentioned above, ripple refers to periodic variation exhibited by a DC voltage. Therefore, circuit **100** can include a combination of components designed to reduce ripple of a Vout **190** at output terminal **185** and, particularly, to ensure that a Vout ripple is less than PCP-Vout ripple. PCP-Vout ripple refers to the difference between the maximum voltage level exhibited by PCP-Vout **119** (i.e., PCP-Vouth) and the minimum voltage level exhibited by PCP-Vout **119** (i.e., PCP-Voutl). aVout ripple refers to the difference between the maximum voltage level exhibited by a Vout **190** (i.e., a Vouth) at output terminal **185** and the minimum voltage level exhibited by aVout **190** (i.e., aVoutl). The ripple reducing components can include a capacitive load **183**, which is electrically connected to an output terminal **185**. These components can also include a pass transistor **181**, which is electrically connected between output node **116** of PCP stage **110** and the output terminal **185** and which is controlled by a supplementary charge pump (SCP) stage **150** connected in parallel with the PCP stage **110**.

[0027] Specifically, SCP stage **150** can include an additional input node **155**, an additional output node **156**, and a pair of additional cross-coupled inverters **151** and **152** connected between additional input node **155** and additional output node **156**. Additional inverter **151** can include an additional NFET **151***a* and an additional PFET **151***b* connected in series between additional input node **155** and additional output node **156**. Additional inverter **152** can similarly include an additional NFET **152***a* and an additional PFET **152***b* connected in series between additional input node **155** and additional output node **156**. An additional intermediate node **157** at the junction between additional NFET **151***a* and additional PFET **151***b* (e.g., on an interconnect electrically connecting additional NFET **151***a* to additional PFET **151***b*) can be electrically connected to the gates of additional NFET **152***a* and additional PFET **152***b*. Similarly, an additional intermediate node **158** at the junction between additional NFET **152***a* and additional PFET **152***b* (e.g., on an interconnect electrically connecting additional NFET **152***a* and additional PFET **152***b*) can be electrically connected to the gates of additional NFET **151***a* and additional PFET **151***b*.

[0028] SCP stage **150** can further include a pair of additional clock signal nodes **161** and **162**. The additional clock signal nodes **161** and **162**, like clock signal nodes **121** and **122**, can be electrically connected to receive CLK**1** and CLK**1**B from clock driver **105**.

[0029] SCP stage **150** can further include a pair of additional capacitors **153** and **154** connected between the additional clock signal nodes **161** and **162** and additional intermediate nodes **157** and **158**, respectively. That is, additional capacitor **153** can include conductive plates, which are connected to additional clock signal node **161** and additional intermediate node **157**, respectively, and which are separated from each other by a capacitor dielectric. Additional capacitor **154** can similarly include conductive plates, which are connected to additional clock signal node **162** and additional intermediate node **158**, respectively, and which are also separated from each other by a capacitor dielectric. It should be noted that the additional capacitors **153** and **154** of SCP stage **150** can be significantly smaller than the capacitors **113** and **114** of PCP stage **110**. For example, capacitance (C) of capacitors **113** and **114** could be **100** times greater than capacitance (Cr1) of additional capacitors **153** and **154**. For example, in some embodiments, C could be 10 picofarads (pF) and Cr1 could be 0.1 pF.

[0030] The additional input node **155** of this SCP stage **150** can also be connected to receive Vin and SCP stage **150** can operate in parallel with and in essentially the same manner as PCP stage **110** in order to output a SCP output voltage (SCP-Vout) **159** at additional output node **156**. SCP stage **150** can further include an additional capacitive load **165** electrically connected to additional output node **156**. Due to the relatively low capacitances of the additional capacitors **153** and **154** and due to additional capacitive load **165**, SCP-Vout **159** (which is applied to gate of pass transistor **181**) exhibits significantly reduced ripple as compared to PCP-Vout **119**. That is, SCP-Vout ripple refers to the difference between the maximum voltage level exhibited by SCP-Vout (also referred to herein as SCP-Vout high (PCP-Vouth)) and the minimum voltage level exhibited by SCP-Vout (also referred to herein as SCP-Vout low (SCP-Voutl) and SCP- Vout ripple is less than PCP-Vout ripple. FIGS. **2**A and **2**B are graphs illustrating and comparing example values for PCP-Vout **119** and SCP-Vout **159** in a circuit **100** where all transistors contained therein are 1.8V transistors and where VDDA is 1.8V.

[0031] The gate of pass transistor **181** can be electrically connected to receive SCP-Vout **159**. Pass transistor **181** can be another NFET such that SCP-Vout **159** continuously maintains pass transistor **181** in an on-state. Thus, pass transistor **181** and capacitive load **183**, in combination, form a low pass filter that reduces ripple. Those skilled in the art will recognize that inclusion of pass transistor **181** circuit **100** will also slightly reduce aVout **190** by some amount (e.g., by approximately 100 mV). Thus, for example, in an embodiment where VDDA is 1.8V and a resistive load **182** is included in the circuit **100**, aVout **190** at output terminal **185** may be pulled down toward 2.9V.

[0032] It should be understood that, during design of a circuit **100** with an SCP stage **150**, the size of CL **183** can be selected to achieve a desired balance between an acceptable a Vout ripple (i.e., the amount of ripple exhibited by aVout **190** at output terminal **185**) and the overall size of circuit **100**. FIG. **2**C is a graph illustrating aVout **190** in a first example (Example 1) of a circuit **100** with design specifications including a set resistive load (RL) **182** (e.g., of 1.6 kilohms (kΩ) for current driving purposes and a set CL **183** (e.g., of 160 picoFarads (pF)). In this case, SCP stage **150** can be incorporated into circuit **100** to further reduce a Vout ripple by, for example, up to 5 times (e.g., from 147 mV to 27 mV). FIG. **2**D is a graph illustrating aVout **190** in a second example (Example 2) of a circuit **100** with design specifications including a set RL **182** (e.g., of 1.6 kΩ) for current driving purposes and an acceptable a Vout ripple of, for example, 147 mV. In this case, a SCP stage **150** can be incorporated into circuit **100** in order to keep aVout ripple at or below 147 mV, while CL **183** is reduced, for example, by approximately 90%. (e.g., from 160 pF to 16 pF). It should be understood that the values provided for RL, CL, aVout ripple in the examples shown in FIGS. **2**A-**2**D are provided for illustration purposes and are not intended to be limiting.

[0033] Referring to FIG. **3**, also disclosed herein are embodiments of a structure and, particularly, of a charge pump circuit **300** (hereinafter referred to as circuit **300**) that includes multiple PCP stages connected in series. This circuit **300** can, like the circuit **100** of FIG. **1**, also include a pass transistor **381** and SCP stage **350**. More particularly, for purposes of illustration, circuit **300** is shown in FIG. **3** as including three PCP stages **310**(**1**)-**310**(**3**) connected in series. However, FIG. **3** is not intended to be limiting. Alternatively, such a circuit **300** could include any number (n) of two or more PCP stages including at least a first PCP stage (e.g., **310**(**1**)) and a last PCP stage (e.g., **310**(**3**)). Additionally, the discussion below refers to a next-to-last PCP stage or a preceding primary charge pump stage, which is immediately adjacent and upstream of the last primary charge pump stage. In a circuit **300** with three PCP stages, as illustrated, the next-to-last PCP stage or preceding PCP stage refers to the second PCP stage **310**(**2**). However, it should be understood that in a circuit with only two PCP stages, this next-to-last PCP stage or preceding PCP stage, which is immediately adjacent and upstream of the last PCP stage would be the first PCP stage. In a circuit with four PCP stages, this next-to-last PCP stage or preceding PCP stage, which is immediately adjacent and upstream of the last PCP stage would be the third PCP stage, and so on.

[0034] Each PCP stage (e.g., **310**(**1**)-**310**(**3**)) can include an input node **315**, an output node **316**, and a pair of cross-coupled inverters **311** and **312** connected between input node **315** and output node **316**. One inverter **311** can include an NFET **311***a* and PFET **311***b* connected in series between input node **315** and output node **316**. Another inverter **312** can similarly include an NFET **312***a* and a PFET **312***b* connected in series between input node **315** and output node **316**. An intermediate node **317** at the junction between NFET **311***a* and PFET **311***b* (e.g., on an interconnect electrically connecting NFET **311***a* to PFET **311***b*) can be electrically connected to the gates of NFET **312***a* and PFET **312***b*. Similarly, an intermediate node **318** at the junction between NFET **312***a* and PFET **312***b* (e.g., on an interconnect electrically connecting NFET **312***a* and PFET **312***b*) can be electrically connected to the gates of NFET **311***a* and PFET **311***b*.

[0035] Each PCP stage (e.g., **310**(**1**)-**310***n*) can further include a pair of clock signal nodes **321** and **322**. Each PCP stage **110** can further include a pair of capacitors **313** and **314** connected between the clock signal nodes **321** and **322** and the intermediate nodes **317** and **318**, respectively. That is, capacitor **313** can include conductive plates, which are connected to clock signal node **321** and intermediate node **117**, respectively, and which are separated from each other by a capacitor dielectric. Capacitor **314** can similarly include conductive plates, which are connected to clock signal node **322** and intermediate node **318**, respectively, and which are also separated from each other by a capacitor dielectric.

[0036] Generally, in operation, first PCP stage **310**(**1**) input node **315** receives Vin**1**, which is steady at VDDA. Due to clock and inverted clock signals on the first PCP stage clock signal nodes **321**-**322** in combination with the capacitances of the first PCP stage capacitors **313** and **314**, the first PCP stage output node **316** outputs PCP**1**-Vout that is continuously pulled toward 2*VDDA. Second PCP stage **310**(**2**) input node **315** receives PCP**1**-Vout. Due to clock and inverted clock signals on the second PCP stage clock signal nodes **321**-**322** in combination with the capacitances of the second PCP stage capacitors **313** and **314**, the second PCP stage output node **316** outputs PCP**2**-Vout that is continuously pulled toward 3*VDDA. Third PCP stage **310**(**3**) input node **315** receives PCP**2**-Vout. Due to clock and inverted clock signals on the third PCP stage clock signal nodes **321**-**322** in combination with the capacitances of the third PCP stage capacitors **313** and **314**, the third PCP stage output node **316** outputs PCP**3**-Vout that is pulled toward 4*VDDA. In other words, the input node of the first PCP stage receives Vin and the input nodes of all downstream stages receive the output voltage from the immediately preceding upstream stage. For example, the last input node of the last PCP stage (e.g., **310**(**1**)) receives that output voltage from the immediately preceding upstream PCP stage (e.g., **310**(**2**)).

[0037] In some embodiments, clock signal nodes **321** in each of the PCP stages can be electrically connected to receive the same clock signal (CLK**1**) and clock signal nodes **322** in each of the PCP stages can be electrically connected to receive the same inverted clock signal (CLK**1**B). For example, if Vin at the input node **315** of the first PCP stage **310**(**1**) is at VDDA, then CLK**1** can swing periodically between 0.0V and VDDA and CLK**1**B can swing periodically between VDDA and 0.0V. That is, as CLK**1** changes from VDDA to 0.0V, CLK**1**B switches from 0.0V to VDDA and vice versa. CLK**1** and CLK**1**B can be generated, for example, by a clock driver that is configured essentially the same as the clock driver **105** of FIG. **1**B described above. In these embodiment, sizes of the capacitors **313** and **314** increase from stage to stage in order to achieve the desired output voltage. For example, if capacitors **313** and **314** in first PCP stage **310**(**1**) each have a capacitance (C), then capacitors **313** and **314** in second PCP stage **310**(**2**) can have a capacitance of 2*C and capacitors **313** and **314** in third PCP stage **310**(**3**) can have a capacitance of 3*C.

[0038] In other embodiments, capacitors **313** and **314** can be the same size so as to have the same capacitance (C) in each of the PCP stages. In these embodiments, the clock and inverter clock signals applied to the clock signal nodes **321** in each of the PCP stages must be different and, particularly, voltage level shifter. For example, clock signal nodes **321** and **322** of first PCP stage **310**(**1**) can receive a first clock signal (CLK**1**) and an inverted first clock signal (CLK**1**B). Clock signal nodes **321** and **322** of second PCP stage **310**(**2**) can receive a second clock signal (CLK**2**) and an inverted second clock signal (CLK**2**B). Clock signal nodes **321** and **322** of third PCP stage **310**(**3**) can receive a third clock signal (CLK**3**) and an inverted third clock signal (CLK**3**B). In these embodiments, if Vin at the input node **315** of the first PCP stage **310**(**1**) is at VDDA, then CLK**1** can swing periodically between 0.0V and VDDA and CLK**1**B can swing periodically between VDDA and 0.0V. That is, as CLK**1** changes from VDDA to 0.0V, CLK**1**B switches from 0.0V to VDDA and vice versa. CLK**2** can swing periodically between VDDA and 2*VDDA and CLK**2**B can swing periodically between 2*VDDA and VDDA. That is, as CLK**2** changes from 2*VDDA to VDDA, CLK**2**B switches from VDDA to 2*VDDA and vice versa. CLK**3** can swing periodically between 2*VDDA and 3*VDDA and CLK**3**B can swing periodically between 3*VDDA and 2*VDDA. That is, as CLK**3** changes from 3*VDDA to 2*VDDA, CLK**3**B switches from 2*VDDA to 3*VDDA and vice versa. Such clock signals can for example be generated by a clock generation circuit that includes a clock driver as well as voltage level shifters to achieve the desired clock and inverted clock signal.

[0039] It should be understood that each PCP stage (e.g., **310**(**1**)-**310**(**3**)) operates in essentially the same manner as PCP stage **110** of FIG. **1**A discussed above, except that the output voltage at the output node various as a function of the input voltage at the input node, voltage levels of the clock and inverter clock signals at the clock signal nodes **321**-**322** and the capacitances of the capacitors

313-314.

[0040] Optionally, circuit **300** can also include a resistive load **382**, which is electrically connected to output terminal **385** for driving current. This resistive load **382** may reduce the voltage level at output terminal **385** somewhat. For example, given a resistive load **382**, an adjusted output voltage (aVout) **390** at output terminal **385** may be pulled down. For example, if PCP**3**-Vout on output node **316** of the last PCP stage **310**(**3**) is on average at a voltage level of 4*VDDA, then aVout **390** at output terminal **185** may be average less than 4*VDDA.

[0041] Furthermore, ripple of PCP**3**-Vout may be significant. Therefore, circuit **300** can include a combination of components designed to reduce ripple of aVout **390** at output terminal **385** and, particularly, to ensure that a Vout ripple is less than PCP**3**-Vout ripple. PCP-Vout ripple refers to the difference between the maximum voltage level exhibited by PCP**3**-Vout and the minimum voltage level exhibited by PCP**3**-Vout. aVout ripple refers to the difference between the maximum voltage level exhibited by aVout **390** at output terminal **385** and the minimum voltage level exhibited by aVout **390**. The ripple reducing components can include a capacitive load **383**, which is electrically connected to an output terminal **385**. These components can also include a pass transistor **381**, which is electrically connected between output node **316** of the last PCP stage (e.g., **310**(**3**)) and the output terminal **385** and which is controlled by a supplementary charge pump (SCP) stage **350** connected in parallel with the last PCP stage.

[0042] Specifically, SCP stage **350** can include an additional input node **355**, an additional output node **356**, and a pair of additional cross-coupled inverters **351** and **352** connected between additional input node **355** and additional output node **356**. Additional inverter **351** can include an additional NFET **351***a* and an additional PFET **351***b* connected in series between additional input node **355** and additional output node **356**. Additional inverter **352** can similarly include an additional NFET **352***a* and an additional PFET **352***b* connected in series between additional input node **355** and additional output node **356**. An additional intermediate node **357** at the junction between additional NFET **351***a* and additional PFET **351***b* (e.g., on an interconnect electrically connecting additional NFET **351***a* to additional PFET **351***b*) can be electrically connected to the gates of additional NFET **352***a* and additional PFET **352***b*. Similarly, an additional intermediate node **358** at the junction between additional NFET **352***a* and additional PFET **352***b* (e.g., on an interconnect electrically connecting additional NFET **352***a* and additional PFET **352***b*) can be electrically connected to the gates of additional NFET **351***a* and additional PFET **351***b*.

[0043] SCP stage **350** can further include a pair of additional clock signal nodes **361** and **362**. The additional clock signal nodes **361** and **362** can be electrically connected to receive the same clock and inverted clock signals as the last PCP stage (e.g., **310**(**3**)). For example, in some embodiments, additional clock signal nodes **361** and **362** can be electrically connected to receive CLK**1** and CLK**1**B (e.g., if all PCP stages receive the same clock and inverter clock signals). In other embodiments, additional clock signal nodes **361** and **362** are electrically connected to receive CLK**3** and CLK**3**B (e.g., if the last PCP stage receives CLK**3** and CLK**3**B).

[0044] SCP stage **350** can further include a pair of additional capacitors **353** and **354** connected between the additional clock signal nodes **361** and **362** and additional intermediate nodes **357** and **358**, respectively. That is, additional capacitor **353** can include conductive plates, which are connected to additional clock signal node **361** and additional intermediate node **357**, respectively, and which are separated from each other by a capacitor dielectric. Additional capacitor **354** can similarly include conductive plates, which are connected to additional clock signal node **362** and additional intermediate node **358**, respectively, and which are also separated from each other by a capacitor dielectric. It should be noted that the additional capacitors **353** and **354** of SCP stage **350** can be significantly smaller than the capacitors **313** and **314** of the last PCP stage (e.g., **310**(**3**)). For example, capacitance (C) of capacitors **313** and **314** of last PCP stage could be one-hundred times greater than capacitance (Cr1) of additional capacitors **353** and **354**.

[0045] The additional input node **355** of this SCP stage **350** can be connected to receive the same input voltage as the last PCP stage (e.g., **310**(**3**)). As mentioned above, the last PCP stage receives the output voltage from the immediately preceding upstream PCP stage (e.g., PCP**2**-Vout from second PCP stage **310**(**2**)). SCP stage **350** can operate in parallel with and in essentially the same manner as the last PCP stage (e.g., **310**(**3**)) in order to output a SCP output voltage (SCP-Vout) **359** at additional output node **356**.

[0046] SCP stage **350** can further include an additional capacitive load **365** electrically connected to additional output node **356**. Due to the relatively low capacitances of the additional capacitors **353** and **354** and due to additional capacitive load **365**, SCP-Vout **359** (which is applied to gate of pass transistor **381**) exhibits significantly reduced ripple as compared to PCP**3**-Vout. The gate of pass transistor **381** can be electrically connected to receive SCP-Vout **359**. Pass transistor **381** can be another NFET such that SCP-Vout **359** continuously maintains pass transistor **381** in an on-state. Thus, pass transistor **381** and capacitive load **383**, in combination, form a low pass filter that reduces ripple. Those skilled in the art will recognize that inclusion of pass transistor **381** circuit **300** will also slightly reduce a Vout **359** by some amount (e.g., by approximately 100 mV).

[0047] It should be understood that, during design of a circuit **300** with an SCP stage **350**, the size of CL **383** can be selected to achieve a desired balance between an acceptable a Vout ripple and the overall size of circuit **300** (e.g., in essentially the same manner as described above with regard to circuit **100**).

[0048] In the method and structures described above, a semiconductor material refers to a material whose conducting properties can be altered by doping with an impurity. Exemplary semiconductor materials include, for example, silicon-based semiconductor materials (e.g., silicon, silicon germanium, silicon germanium carbide, silicon carbide, etc.) and III-V compound semiconductors (i.e., compounds obtained by combining group III elements, such as aluminum (Al), gallium (Ga), or indium (In), with group V elements, such as nitrogen (N), phosphorous (P), arsenic (As) or antimony (Sb)) (e.g., GaN, InP, GaAs, or GaP). A pure semiconductor material and, more particularly, a semiconductor material that is not doped with an impurity for the purposes of increasing conductivity (i.e., an undoped semiconductor material) is referred to in the art as an intrinsic semiconductor. A semiconductor material that is doped with an impurity for the purposes of increasing conductivity (i.e., a doped semiconductor material) is referred to in the art as an extrinsic semiconductor and will be more conductive than an intrinsic semiconductor made of the same base material. That is, extrinsic silicon will be more conductive than intrinsic silicon; extrinsic silicon germanium will be more conductive than intrinsic silicon germanium; and so on. Furthermore, it should be understood that different impurities (i.e., different dopants) can be used to achieve different conductivity types (e.g., P-type conductivity and N-type conductivity) and that the dopants may vary depending upon the different semiconductor materials used. For example, a silicon-based semiconductor material (e.g., silicon, silicon germanium, etc.) is typically doped with a Group III dopant, such as boron (B) or indium (In), to achieve P-type conductivity, whereas a silicon-based semiconductor material is typically doped with a Group V dopant, such as arsenic (As), phosphorous (P) or antimony (Sb), to achieve N-type conductivity. A gallium nitride (GaN)-based semiconductor material is typically doped with magnesium (Mg) to achieve P-type conductivity and with silicon (Si) or oxygen to achieve N-type conductivity. Those skilled in the art will also recognize that different conductivity levels will depend upon the relative concentration levels of the dopant(s) in a given semiconductor region.

[0049] It should be understood that the terminology used herein is for the purpose of describing the disclosed structures and methods and is not intended to be limiting. For example, as used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Additionally, as used herein, the terms “comprises,” “comprising,” “includes,” and/or “including” specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

Furthermore, as used herein, terms such as “right,” “left,” “vertical,” “horizontal,” “top,” “bottom,” “upper,” “lower,” “under,” “below,” “underlying,” “over,” “overlying,” “parallel,” “perpendicular,” etc., are intended to describe relative locations as they are oriented and illustrated in the drawings (unless otherwise indicated) and terms such as “touching,” “in direct contact,” “abutting,” “directly adjacent to,” “immediately adjacent to,” etc., are intended to indicate that at least one element physically contacts another element (without other elements separating the described elements). The term “laterally” is used herein to describe the relative locations of elements and, more particularly, to indicate that an element is positioned to the side of another element as opposed to above or below the other element, as those elements are oriented and illustrated in the drawings. For example, an element that is positioned laterally adjacent to another element will be beside the other element, an element that is positioned laterally immediately adjacent to another element will be directly beside the other element, and an element that laterally surrounds another element will be adjacent to and border the outer sidewalls of the other element. The corresponding structures, materials, acts, and equivalents of all means or step plus function elements in the claims below are intended to include any structure, material, or act for performing the function in combination with other claimed elements as specifically claimed.
[0050] The method as described above is used in the fabrication of integrated circuit chips. The resulting integrated circuit chips can be distributed by the fabricator in raw wafer form (that is, as a single wafer that has multiple unpackaged chips), as a bare die, or in a packaged form. In the latter case the chip is mounted in a single chip package (such as a plastic carrier, with leads that are affixed to a motherboard or other higher level carrier) or in a multichip package (such as a ceramic carrier that has either or both surface interconnections or buried interconnections). In any case the chip is then integrated with other chips, discrete circuit elements, and/or other signal processing devices as part of either (a) an intermediate product, such as a motherboard, or (b) an end product. The end product can be any product that includes integrated circuit chips, ranging from toys and other low-end applications to advanced computer products having a display, a keyboard or other input device, and a central processor.
[0051] The descriptions of the various disclosed embodiments have been presented for purposes of illustration but are not intended to be exhaustive or limiting. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the disclosed embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

Claims

1. A structure comprising: a primary charge pump stage including an input node and an output node; supplementary charge pump stage including an additional input node and an additional output node, wherein the input node and the additional input node are connected to receive an input voltage; an output terminal; and a pass transistor connected between the output terminal and the output node and having a gate connected to the additional output node.
2. The structure of claim 1, wherein the primary charge pump stage includes: cross-coupled inverters connected between the input node and the output node, wherein the cross-coupled inverters have intermediate nodes, respectively, with each intermediate node being at an interconnect between a P-channel field effect transistor (PFET) and an N-channel field effect transistor (NFET) of one of the cross-coupled inverters; clock signal nodes connected to receive clock and inverted clock signals, respectively; and capacitors connected between the clock signal nodes and the intermediate nodes.
3. The structure of claim 2, wherein the supplementary charge pump stage includes: additional

cross-coupled inverters connected between the additional input node and the additional output node, wherein the additional cross-coupled inverters have additional intermediate nodes, respectively, with each additional intermediate node being at an interconnect between an additional PFET and an additional NFET of one of the additional cross-coupled inverters; additional clock signal nodes connected to receive the clock and inverted clock signals, respectively; and additional capacitors connected between the additional clock signal nodes and the additional intermediate nodes.
4. The structure of claim 3, wherein the capacitors are larger than the additional capacitors.
5. The structure of claim 3, wherein the capacitors are at least 100 times larger than the additional capacitors.
6. The structure of claim 1, wherein the output terminal is connected to a capacitive load.
7. The structure of claim 1, wherein the output terminal is connected to a resistive load.
8. A structure comprising: a primary charge pump stage including an input node and an output node; a supplementary charge pump stage including an additional input node and an additional output node, wherein the input node and the additional input node are connected to receive an input voltage; an output terminal connected to a capacitive load; and a pass transistor connected between the output terminal and the output node and having a gate connected to the additional output node.
9. The structure of claim 8, wherein the primary charge pump stage includes: cross-coupled inverters connected between the input node and the output node, wherein the cross-coupled inverters have intermediate nodes, respectively, with each intermediate node being at an interconnect between a P-channel field effect transistor (PFET) and an N-channel field effect transistor (NFET) of one of the cross-coupled inverters; clock signal nodes connected to receive clock and inverted clock signals, respectively; and capacitors connected between the clock signal nodes and the intermediate nodes.
10. The structure of claim 9, wherein the supplementary charge pump stage includes: additional cross-coupled inverters connected between the additional input node and the additional output node, wherein the additional cross-coupled inverters have additional intermediate nodes, respectively, with each additional intermediate node being at an interconnect between an additional PFET and an additional NFET of one of the additional cross-coupled inverters; additional clock signal nodes connected to receive the clock and inverted clock signals, respectively; and additional capacitors connected between the additional clock signal nodes and the additional intermediate nodes.
11. The structure of claim 10, wherein the capacitors are larger than the additional capacitors.
12. The structure of claim 10, wherein the capacitors are at least 100 times larger than the additional capacitors.
13. The structure of claim 8, wherein the output terminal is connected to a resistive load.
14. A structure comprising: multiple primary charge pump stages connected in series including at least a first primary charge pump stage and a last primary charge pump stage, wherein the last primary charge pump stage of the multiple primary charge pump stages includes a last input node and a last output node; a supplementary charge pump stage including an additional input node and an additional output node, wherein the last input node and the additional input node are connected to receive a last input voltage from a preceding primary charge pump stage; an output terminal; and a pass transistor connected between the output terminal and the last output node and having a gate connected to the additional output node.
15. The structure of claim 14, wherein each primary charge pump stage includes: cross-coupled inverters connected between input and output nodes, wherein the cross-coupled inverters have intermediate nodes, respectively, with each intermediate node being at an interconnect between a P-channel field effect transistor (PFET) and an N-channel field effect transistor (NFET) of one of the cross-coupled inverters; clock signal nodes connected to receive clock and inverted clock signals, respectively; and capacitors connected between the clock signal nodes and the intermediate nodes.

16. The structure of claim 15, wherein the supplementary charge pump stage includes: additional cross-coupled inverters connected between the additional input node and the additional output node, wherein the additional cross-coupled inverters have additional intermediate nodes, respectively, with each additional intermediate node being at an interconnect between an additional PFET and an additional NFET of one of the additional cross-coupled inverters; additional clock signal nodes connected to receive the clock and inverted clock signals, respectively; and additional capacitors connected between the additional clock signal nodes and the additional intermediate nodes.

17. The structure of claim 16, wherein the capacitors of the last primary charge pump stage are larger than the additional capacitors of the supplementary charge pump stage.

18. The structure of claim 16, wherein the capacitors of the last primary charge pump stage are at least 100 times larger than the additional capacitors of the supplementary charge pump stage.

19. The structure of claim 14, wherein the output terminal is connected to a capacitive load.

20. The structure of claim 14, wherein the output terminal is connected to a resistive load.